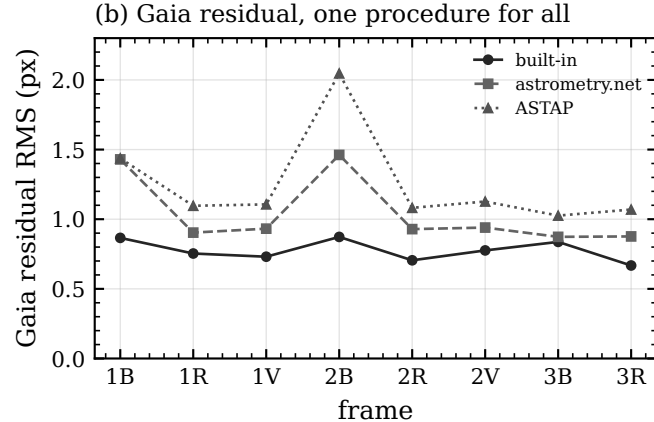
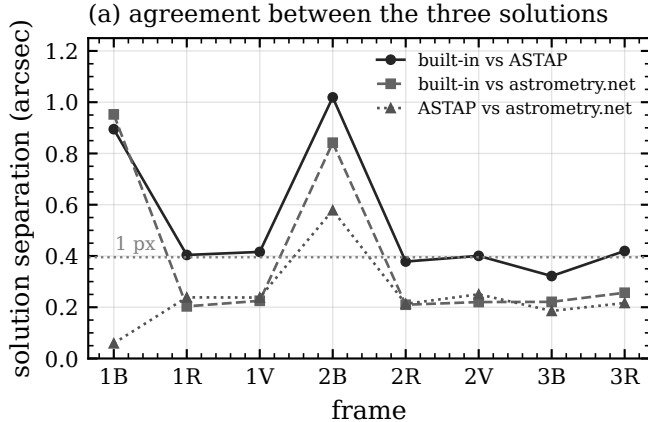




Real data: M13, eight 60 s single exposures (B 1-3, R 1-3, V 1-2), Moravian C3-61000, bin 2, 0.395"/px, 2026-05-15.

One copy tree per engine, the same Step-4 detection list shared by all three; engines: built-in quad solver, ASTAP, astrometry.net. Source: validation/paper/data_wcs_engines (run_wcs_engine_cross.py).